

SARYG-SEP, Russian Federation

WMO: 361040

Lat: 51.4789N

Lon: 95.5778E

Elev: 2316

StdP: 13.51

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.2	(c) -39.8	(d) -48.8	(e) 0.3	(f) -44.3	(g) -44.5	(h) 0.5	(i) -39.9	(j) 8.1	(k) -12.7	(l) 7.7	(m) -12.7	(n) 0.5	(o) 120	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	89.9	63.8	85.5	62.5	81.8	61.5	66.7	82.4	65.3	79.6	63.9	77.2	4.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	90.9	69.9	60.7	86.3	68.6	59.2	81.7	67.9	32.7	81.7	31.5	79.9	30.5	77.5	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.9	(b) 9.7	(c) 8.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 11.9	(5) 9.7	(6) 8.2	(7) -47.5	(8) 69.5	(9) 3.6	(10) 1.8	(11) -50.1	(12) 70.7	(13) -52.2	(14) 71.8	(15) -54.2	(16) 72.8	(17) -56.7	(18) 74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	-21.4	-11.4	15.9	40.4	49.8	63.4	66.9	63.1	50.5	34.7	8.2	-12.8		
	DBStd	32.16	9.07	11.06	13.86	8.04	8.20	6.93	5.03	5.80	7.76	8.59	13.36	10.60		
	HDD50	9164	2214	1719	1056	299	107	2	0	0	88	478	1254	1947		
	HDD65	13291	2679	2139	1521	739	472	110	35	104	436	940	1704	2412		
	CDD50	1555	0	0	0	10	102	405	525	407	103	3	0	0		
	CDD65	206	0	0	0	0	2	62	95	46	1	0	0	0		
	CDH74	3370	0	0	0	8	189	1161	1280	656	76	0	0	0		
	CDH80	1184	0	0	0	0	45	441	461	227	10	0	0	0		
	WSAvg	3.9	2.6	3.4	4.1	4.7	5.1	4.5	4.4	4.0	4.2	3.8	3.2	2.8		
	PrecAvg	13.1	0.4	0.2	0.3	0.6	1.0	1.9	2.9	2.5	1.4	0.7	0.7	0.6		
(15) Precipitation	PrecMax	16.9	0.7	0.6	0.7	1.1	1.7	2.6	5.0	4.1	2.5	1.5	1.4	1.0		
	PrecMin	8.9	0.2	0.0	0.1	0.2	0.5	1.0	1.3	1.0	0.6	0.2	0.3	0.2		
	PrecStd	2.0	0.2	0.1	0.2	0.2	0.4	0.5	1.0	0.9	0.6	0.3	0.3	0.2		
	PrecStd	2.0	0.2	0.1	0.2	0.2	0.4	0.5	1.0	0.9	0.6	0.3	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	23.2	51.1	74.9	86.9	93.5	96.1	92.3	81.3	66.1	42.4	22.3		
		MCWB	9.8	21.8	38.5	51.3	57.3	64.5	66.2	65.3	58.9	47.2	33.7	21.1		
	2%	DB	1.8	14.8	45.1	68.0	80.0	89.1	90.2	85.8	75.9	58.7	35.2	12.7		
		MCWB	1.1	13.3	36.2	49.0	55.9	63.0	65.2	63.9	57.7	45.1	31.0	11.7		
	5%	DB	-2.3	10.9	40.3	61.9	74.6	85.2	84.8	80.8	70.4	52.5	30.6	7.3		
		MCWB	-3.1	9.4	33.4	45.3	53.3	61.9	63.4	63.0	55.5	42.2	28.5	6.4		
	10%	DB	-6.5	7.1	35.5	57.1	69.7	81.1	81.1	76.3	65.0	47.6	26.9	2.5		
		MCWB	-7.2	6.0	31.0	42.7	51.4	60.5	62.6	61.2	52.9	39.9	25.1	1.8		
	0.4%	WB	9.8	21.3	39.8	53.2	60.9	68.4	68.6	68.7	61.0	49.4	35.0	21.1		
		MCDB	10.2	23.5	49.5	69.7	80.7	86.6	86.9	85.7	75.8	63.8	39.7	22.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.3	13.4	36.5	49.4	57.2	66.1	66.7	66.1	58.9	46.0	31.2	11.9		
		MCDB	1.8	14.6	44.1	64.8	75.0	83.1	81.8	80.5	72.9	56.5	34.5	12.9		
	5%	WB	-3.0	9.6	33.8	46.0	54.9	64.4	65.4	64.5	56.8	43.0	28.5	6.5		
		MCDB	-2.2	10.9	39.2	59.5	71.7	79.1	79.1	77.3	67.8	51.8	30.7	7.2		
	10%	WB	-7.0	6.0	30.9	43.6	52.0	62.4	64.2	62.6	53.9	40.1	25.2	2.0		
		MCDB	-6.4	7.1	35.7	56.0	67.5	77.2	77.5	73.6	63.4	46.4	26.9	2.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	24.9	25.4	25.4	27.2	28.3	25.2	24.2	26.0	21.2	18.2	15.9		
		MCDBR	19.8	24.2	22.7	34.2	38.2	36.8	33.0	33.0	34.7	29.9	18.9	16.0		
		MCWBR	19.1	22.7	14.4	18.3	18.6	14.9	12.6	14.6	16.7	17.0	14.5	15.3		
	5% WB	MCDBR	20.1	24.3	21.9	30.1	33.8	31.3	26.5	27.6	31.2	28.9	18.1	15.8		
		MCWBR	19.5	22.9	14.1	17.2	17.7	14.4	11.3	13.5	16.3	16.8	14.2	15.2		
			(d) 19.6	(e) 24.9	(f) 25.4	(g) 25.4	(h) 27.2	(i) 28.3	(j) 25.2	(k) 24.2	(l) 26.0	(m) 21.2	(n) 18.2	(o) 15.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.296	0.343	0.386	0.407	0.393	0.396	0.380	0.336	0.320	0.299	0.271		
	taud		2.128	2.081	2.056	2.195	2.176	2.326	2.349	2.362	2.432	2.376	2.215	2.097		
	Ebn at Noon		233	257	265	269	268	273	271	269	271	250	217	205		
	Edh at Noon		23	33	42	41	44	38	37	35	29	25	22	21		
(44) All-Sky Solar Radiation	RadAvg		435	798	1331	1609	1838	2024	1858	1599	1230	765	416	320		
	RadStd		17	30	45	79	107	138	108	93	111	50	32	14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page